

Nanyang Technological University
SPMS/Division of Mathematical Sciences

AY20/21 MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS Test 1

Name &
Matriculation Number

Score:

Venue:

TIME: 9:35 -10:15, 15 September, 2020

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1. (40 Points) A factory finds that, on average, 20% of the bolts produced by a given machine will be defective for certain specified requirements. If 10 bolts are selected at random from the day's production of this machine, find the probability
 - (a) that exactly 2 will be defective;
 - (b) that 2 or more will be defective.

2. (30 Points) A laboratory blood test is 95% effective in detecting a ceratin disease when it is, in fact, present. However, the test also yields a “false positive” result for 1% of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply that he or she has the disease.) If 0.5% of the population actually has the disease, what is the probability that a person has the disease that the test result in positive?

3. (30 Points) The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ . However, such a random variable can be observed only if it is positive, since if it is 0, then we cannot know that such an insect was on the leaf. If we let Y denote the observed number of eggs, then

$$P\{Y = i\} = P\{X = i | X > 0\}$$

where X is Poisson with parameter λ . Find $\mathbb{E}(Y)$.